ENGR 1050
Homework 5: Data Collection using the Google API

Due: November 16th, 2026 by 11:59 PM

Instructions

In class we vibe coded a way to:

1. Make a Google form to collect data
2. Push that data into a Google sheet
3. Use the Google Sheets API to read that data into a Python script
4. Plot that data in Python

The following is a very short homework - its purpose is to crowdsource some data that we will use for machine learning in the next couple weeks.

Problem 1: Collecting Data via Google Forms (50 points)

Generate a Google Form that collects the following data from each respondent:

- State of residence prior to attending Penn
- Preferred fast food, selected from a list of the following options:
  - McDonald's
  - Burger King
  - Wendy's
  - Chick-fil-A
  - Taco Bell
  - In-N-Out
  - Culver's

Ask 5 different people from outside this class to fill out your form. Prioritize asking people from different states to get a more diverse dataset. Take a screenshot of your Google Form once you have created it.

Problem 2: Data Analysis (50 points)

Using the code in class, extract the data into a CSV file with two columns: State and Preferred Fast Food. Then, create a bar chart showing the number of respondents for each fast food option.

Submission

Submit to Canvas:

- A screenshot of your Google Form
- The CSV file containing the collected data, which Prof. Trask will aggregate and share with the class